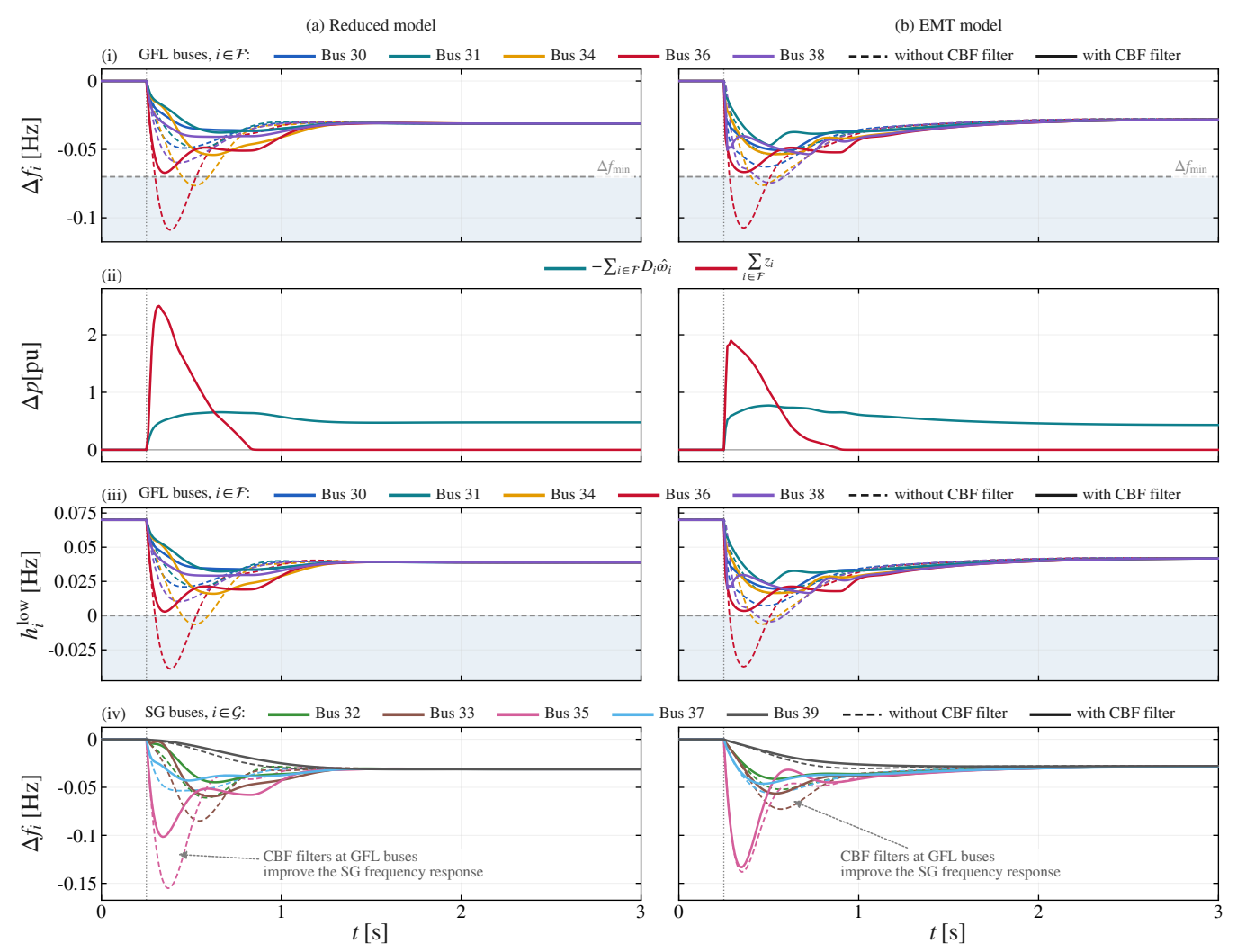

(a) Reduced model
(b) EMT model
(i) GFL buses, $i \in \mathcal{F}$:
Bus 30
Bus 31
Bus 34
Bus 36
Bus 38
without CBF filter
with CBF filter
Δf_i [Hz]
$\Delta f_{\min}$
(ii)
$-\sum_{i\in\mathcal{F}} D_i \hat{\omega}_i$
$\sum_{i\in\mathcal{F}} z_i$
Δp[pu]
(iii) GFL buses, $i \in \mathcal{F}$:
h_i^{low} [Hz]
(iv) SG buses, $i \in \mathcal{G}$:
Bus 32
Bus 33
Bus 35
Bus 37
Bus 39
CBF filters at GFL buses
improve the SG frequency response
t [s]